Dittert's Conjecture in Dimensions 14, 15, and 16 via a Two-Zero Boundary Gap

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Abstract

Let K_n be the set of nonnegative $n \times n$ matrices whose entries sum to n, and let

$$\Phi(A) = \prod_{i=1}^{n} r_i + \prod_{j=1}^{n} c_j - \operatorname{per}(A),$$

where r_i and c_j are the row and column sums. We prove, by a short exact computer-assisted calculation, that Dittert's conjecture holds for $n = 14, 15, 16$:

$$\Phi(A) \le 2 - \frac{n!}{n^n},$$

with equality only at the matrix all of whose entries are $1/n$. The proof combines the joint-deficit scaling lemma with two facts about the boundary: a maximizing matrix must contain two zeroes in distinct rows and columns, and the minimum permanent on that two-zero face of the Birkhoff polytope reduces to a one-variable polynomial. Exact rational certificates for the three polynomial lower bounds and the final scalar comparisons are included with this note.

1 Setup and known results

Let

$$K_n = \left\{ A = (a_{ij}) \in \mathbb{R}_{\ge 0}^{n\times n} : \sum_{i,j} a_{ij} = n \right\}, \qquad \gamma_n = \frac{n!}{n^n},$$

and put $U_n = n^{-1}\mathbf{1}\mathbf{1}^{\mathsf{T}}$. Dittert's conjecture asserts that

$$\Phi(A) \le \Phi(U_n) = 2 - \gamma_n, \tag{1}$$

with equality only at U_n.

We use the following established results.

(i) Every positive global maximizer of Φ on K_n is U_n [2].

(ii) A partly decomposable matrix cannot be a global maximizer of Φ, and the complete zero set of a global maximizer cannot be one rectangular block [1, Theorems 3.2 and 3.7].

(iii) A nonnegative matrix S is doubly superstochastic if and only if

$$s(S[I,J]) \ge |I| + |J| - n \tag{2}$$

for every pair of row and column sets I, J; here s denotes the sum of the indicated submatrix [1, Lemma 2.2].

2 A boundary maximizer has two independent zeroes

Lemma 2.1. *Let Z be a nonempty set of positions in an $n \times n$ array. If Z does not contain two positions in distinct rows and distinct columns, then Z is contained in one row or in one column.*

Proof. Choose $(i_0, j_0) \in Z$. If every element of Z lies in row i_0, there is nothing to prove. Otherwise choose $(i_1, j_0) \in Z$ with $i_1 \neq i_0$; the column must be j_0 by the hypothesis. If $(i, j) \in Z$ had $j \neq j_0$, then it would have to share a row with both (i_0, j_0) and (i_1, j_0), which is impossible. Hence every element of Z lies in column j_0. □

Corollary 2.2. *Every boundary global maximizer of Φ contains two zero entries in distinct rows and distinct columns.*

Proof. By Lemma 2.1, otherwise all zeroes lie in one row or one column. If that row or column is entirely zero, the matrix is partly decomposable. If not, the complete zero set is one proper rectangular block. Both alternatives are excluded by the results quoted in Section 1. □

3 Joint-deficit scaling

The following is the joint-deficit scaling observation used independently for the case $n = 16$ in [3]. We reproduce the proof because the shared deficit budget is central to the extension to $n = 14, 15$.

Lemma 3.1 (Joint-deficit scaling)**.** *Suppose $A \in K_n$ satisfies*

$$\Phi(A) \geq 2 - \gamma_n, \qquad \operatorname{per}(A) = \gamma_n - \delta, \qquad 0 \leq \delta < 1.$$

Set

$$t = \sqrt{\frac{n\delta}{1-\delta}}.$$

If $t < 1$, then $(1-t)^{-1}A$ is doubly superstochastic.

Proof. Write

$$R = \prod_i r_i = 1 - \rho, \qquad C = \prod_j c_j = 1 - \sigma.$$

AM–GM gives $R, C \leq 1$. The assumed lower bound on Φ gives

$$\rho + \sigma \leq \delta. \tag{3}$$

Let I be a set of k rows and write $\sum_{i \in I} r_i = k + \varepsilon$. For $k = 0$ or $k = n$ we have $\varepsilon = 0$, so the required deviation bound is immediate. We may therefore assume $1 \leq k \leq n-1$. Applying AM–GM inside I and its complement gives

$$R \leq \left(1 + \frac{\varepsilon}{k}\right)^k \left(1 - \frac{\varepsilon}{n-k}\right)^{n-k}.$$

With $p = k/n$ and $q = (k+\varepsilon)/n$, the logarithm of the right-hand side is $-nD(p\|q)$. Binary Pinsker, $D(p\|q) \geq 2(p-q)^2$, therefore yields

$$|\varepsilon| \leq \sqrt{-\frac{n}{2}\log(1-\rho)} \leq \sqrt{\frac{n\rho}{2(1-\delta)}}.$$

The same estimate holds for every column subset, with σ in place of ρ.

Now take row and column sets I, J and put $p_0 = |I| + |J| - n > 0$. By inclusion–exclusion and nonnegativity,

$$s(A[I,J]) \geq p_0 - \sqrt{\frac{n\rho}{2(1-\delta)}} - \sqrt{\frac{n\sigma}{2(1-\delta)}}$$
$$\geq p_0 - \sqrt{\frac{n(\rho+\sigma)}{1-\delta}} \geq p_0 - t,$$

where we used (3). Since p_0 is a positive integer, $p_0 - t \geq (1-t)p_0$. Thus $(1-t)^{-1}A$ satisfies (2). Pairs with $p_0 \leq 0$ satisfy it automatically. $\square$

4 The permanent gap on the two-zero face

Let $\Omega_n^{(2)}$ be the set of doubly stochastic $n \times n$ matrices having prescribed zeroes in two distinct rows and columns. Permuting columns, we may take those zeroes to be positions $(1,2)$ and $(2,1)$.

Put $m = n - 2$. In the notation of Pula–Song–Wanless this is the face $\Omega(V_{m,2})$ [6]. Minc's averaging theorem for permanent minimizers with repeated support rows and columns implies that a minimizer on this face may be chosen in the block form [4, Theorem 1]; see also [6]

$$Y = \begin{pmatrix} x_1 & 0 & a_1\mathbf{1}_m^\mathsf{T} \\ 0 & x_2 & a_2\mathbf{1}_m^\mathsf{T} \\ a_1\mathbf{1}_m & a_2\mathbf{1}_m & xJ_m \end{pmatrix}, \qquad x_i = 1 - ma_i, \qquad a_1 + a_2 + mx = 1. \tag{4}$$

Here $0 \leq a_i \leq 1/m$. Only the averaging step of Minc's theorem is used to obtain (4). Section 2 of Pula–Song–Wanless assumes both block parameters exceed 2, so its equalization theorem is not invoked for the present $V_{m,2}$ face. Although Minc already resolved this face, the following short calculation supplies the one-variable reduction directly and avoids importing any unrecorded formula for the minimizer.

Lemma 4.1 (Equalization of the exceptional parameters)**.** *For $m \geq 4$, among matrices of the form* (4) *with fixed $s = a_1 + a_2$, the permanent is minimized when $a_1 = a_2 = s/2$.*

Proof. Put $p = a_1a_2$ and $x = (1-s)/m$. Counting permanent terms according to whether both, one, or neither of the two entries x_1, x_2 are used gives

$$\operatorname{per}(Y) = m!x^{m-2}\Big[x^2x_1x_2 + mx(x_1a_2^2 + x_2a_1^2) + m(m-1)a_1^2a_2^2\Big]. \tag{5}$$

The symmetric identities

$$x_1x_2 = 1 - ms + m^2p, \qquad x_1a_2^2 + x_2a_1^2 = s^2 - (2+ms)p$$

show that the bracket in (5), as a polynomial in p, has derivative

$$2m(m-1)p + (m+1)s^2 - ms - 1. \tag{6}$$

The constraints $0 \leq a_i \leq 1/m$ imply $0 \leq s \leq 2/m$, while $p \leq s^2/4$. Since (6) increases with p, it is at most

$$D_m(s) = \frac{(m^2+m+2)s^2 - 2ms - 2}{2}.$$

The quadratic D_m is convex, and

$$D_m(0) = -1, \qquad D_m(2/m) = -1 + \frac{2}{m} + \frac{4}{m^2} < 0 \quad (m \geq 4).$$

Hence (6) is strictly negative throughout the feasible region. For fixed s, increasing p therefore decreases the permanent, and the largest possible value $p = s^2/4$ is attained exactly at $a_1 = a_2 = s/2$. $\square$

It follows that a minimizing matrix may be chosen as

$$B_m(x) = \begin{pmatrix} a & 0 & b\mathbf{1}_m^\mathsf{T} \\ 0 & a & b\mathbf{1}_m^\mathsf{T} \\ b\mathbf{1}_m & b\mathbf{1}_m & xJ_m \end{pmatrix}, \quad a = \frac{2 - m + m^2 x}{2}, \quad b = \frac{1 - mx}{2}, \tag{7}$$

where

$$\frac{m-2}{m^2} \leq x \leq \frac{1}{m}. \tag{8}$$

A second reading of the three cases in (5) gives

$$\operatorname{per} B_m(x) = m! x^{m-2} \big(x^2 a^2 + 2mxab^2 + m(m-1)b^4 \big). \tag{9}$$

Equivalently,

$$P_n(x) := \operatorname{per} B_m(x) = \frac{(m!)^2}{16} \sum_{i=0}^{2} \binom{2}{i} \frac{2^i}{(m-2+i)!} (2 - m + m^2 x)^i (1 - mx)^{4-2i} x^{m-2+i}. \tag{10}$$

For the three dimensions, exact expansion gives

$$\begin{aligned} P_{14}(x) &= 119750400 x^{10} (953856 x^4 - 289728 x^3 + 33220 x^2 - 1704 x + 33), \\ P_{15}(x) &= 1556755200 x^{11} (1513733 x^4 - 427739 x^3 + 45582 x^2 - 2171 x + 39), \\ P_{16}(x) &= 10897286400 x^{12} (4648336 x^4 - 1227744 x^3 + 122200 x^2 - 5432 x + 91). \end{aligned}$$

Their derivatives factor as

$$\begin{aligned} P_{14}'(x) &= 718502400 x^9 (84x - 5)(26496 x^3 - 5896 x^2 + 440 x - 11), \\ P_{15}'(x) &= 20237817600 x^{10} (195 x - 11)(8957 x^3 - 1857 x^2 + 129 x - 3), \\ P_{16}'(x) &= 305124019200 x^{11} (56 x - 3)(47432 x^3 - 9204 x^2 + 598 x - 13). \end{aligned}$$

The three cubic factors are strictly increasing on $\mathbb{R}$: the discriminants of their derivatives are respectively

$$-847616, \qquad -71640, \qquad -1517568.$$

The linear factors are positive throughout the intervals (8), and the cubic endpoint values are

n	$c_n((m-2)/m^2)$	$c_n(1/m)$
14	$-1/216$	$1/18$
15	$-2/2197$	$2/169$
16	$-1/343$	$2/49$.

Thus each P_n has one global minimum on its admissible interval. Exact rational brackets for the critical points are

$$\begin{aligned} n = 14: &\quad \frac{70587793823672}{10^{15}} < x_* < \frac{70587793823673}{10^{15}}, \\ n = 15: &\quad \frac{65989565762012}{10^{15}} < x_* < \frac{65989565762013}{10^{15}}, \\ n = 16: &\quad \frac{61946716891694}{10^{15}} < x_* < \frac{61946716891695}{10^{15}}. \end{aligned}$$

Exact rational interval-Horner evaluation on these brackets gives

$$\begin{array}{c|c|c} n & \min\limits_{B\in\Omega_n^{(2)}} \operatorname{per}(B) > & L_n \\ \hline 14 & 7890634087/10^{15} & 789/10^8 \\ 15 & 3001151594/10^{15} & 3001/10^9 \\ 16 & 1139154849/10^{15} & 1139/10^9. \end{array} \tag{11}$$

The first verifier supplied with this note checks the displayed factorizations, monotonicity, critical-point brackets, and outward rational interval evaluations using only integer and rational arithmetic. An independent verifier instead applies Sturm's theorem directly to $P_n(x) - L_n$ on the full interval (8).

5 Boundary exclusion

Proposition 5.1. *Fix $n \geq 4$. Suppose every $B \in \Omega_n^{(2)}$ satisfies* $\operatorname{per}(B) \geq L$ *and*

$$\eta := L - \gamma_n - \frac{n^3 L^2}{4(1-\gamma_n)} > 0. \tag{12}$$

Then a global maximizer of Φ on K_n cannot lie on the boundary.

Proof. Let A be a boundary global maximizer. Since $\Phi(A) \geq \Phi(U_n)$ and the two product terms are at most 1, write

$$\operatorname{per}(A) = \gamma_n - \delta, \qquad 0 \leq \delta \leq \gamma_n.$$

If $\delta = 0$, both product bounds are equalities, so A is doubly stochastic; the equality case in the van der Waerden theorem then gives $A = U_n$, contrary to the assumed zero. Hence $0 < \delta \leq \gamma_n$.

By Corollary 2.2, A has two independent zeroes. Set $t = \sqrt{n\delta/(1-\delta)}$. For $n = 14, 15, 16$ we have $(n+1)\gamma_n < 1$, and therefore $t < 1$. Lemma 3.1 gives a doubly stochastic matrix $B \leq (1-t)^{-1}A$. The two zeroes of A are also zeroes of B, so $\operatorname{per}(B) \geq L$. Monotonicity and homogeneity of the permanent give

$$\gamma_n - \delta = \operatorname{per}(A) \geq L(1-t)^n. \tag{13}$$

On the other hand, Bernoulli's inequality and $\delta \leq \gamma_n$ imply

$$\begin{aligned} L(1-t)^n - (\gamma_n - \delta) &\geq L - \gamma_n + \delta - nL\sqrt{\frac{n\delta}{1-\delta}} \\ &\geq L - \gamma_n + x^2 - nL\sqrt{\frac{n}{1-\gamma_n}}\,x \\ &\geq L - \gamma_n - \frac{n^3L^2}{4(1-\gamma_n)} = \eta > 0, \end{aligned}$$

where $x = \sqrt{\delta}$ and the final step completes the square. This contradicts (13). $\square$

For the values in (11), exact rational arithmetic gives

n	η_n
14	$\dfrac{1801777553768587612545302097}{95789864173269226655187500000000000} > 0$
15	$\dfrac{172396552001926445448721}{242137082533381470720000000000000} > 0$
16	$\dfrac{4164538286276424752117963756736721}{1208924448418671074738176000000000000000000} > 0.$

Numerically these margins are approximately $1.88097 \cdot 10^{-9}$, $7.11979 \cdot 10^{-9}$, and $3.44483 \cdot 10^{-9}$, respectively.

Theorem 5.2. *For $n \in \{14, 15, 16\}$ and every $A \in K_n$,*

$$\Phi(A) \le 2 - \frac{n!}{n^n},$$

with equality if and only if $A = U_n$.

Proof. A global maximizer exists by compactness. Proposition 5.1 and the exact checks above exclude a boundary maximizer. Every global maximizer is therefore positive, and Hwang's theorem identifies it uniquely as U_n. □

Corollary 5.3. *Together with Pang's result for $n \ge 17$ [5], Dittert's conjecture holds for every $n \ge 14$.*

6 Reproducibility

The accompanying files are:

- `verify_dittert_n14_n16.py`: a standard-library-only exact verifier. It reconstructs the two-parameter expression (5), verifies Lemma 4.1, reconstructs (10), checks all displayed factorizations, proves the critical-point brackets and interval lower bounds, and verifies (12) as a rational inequality.

- `audit_dittert_n14_n16_sympy.py`: an independent exact audit. It derives the block permanent separately, verifies the equalization step, checks the permanent polynomial by exact Ryser evaluation at $n + 1$ rational points, and uses Sturm root counts to prove $P_n - L_n > 0$ on each complete admissible interval.

Neither program uses a floating-point number in a correctness decision. Both terminate with a certification message for all three dimensions.

References

[1] G.-S. Cheon and I. M. Wanless, *Some results towards the Dittert conjecture on permanents*, Linear Algebra Appl. 436 (2012), 791–801.

[2] S.-G. Hwang, *A note on a conjecture on permanents*, Linear Algebra Appl. 76 (1986), 31–44.

[3] B. Kafidov, *Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma*, arXiv:2607.19439v1 (2026).

[4] H. Minc, *Minimum permanents of doubly stochastic matrices with prescribed zero entries*, Linear and Multilinear Algebra 15 (1984), 225–243.

[5] Z. Pang, *Proof of Dittert's conjecture for dimensions $n \geq 17$*, arXiv:2606.01531v1 (2026).

[6] K. Pula, S.-Z. Song, and I. M. Wanless, *Minimum permanents on two faces of the polytope of doubly stochastic matrices*, Linear Algebra Appl. 434 (2011), 232–238.